

IE4497/EE6497 Pattern Recognition and Deep Learning

Homework 1 — Complete Solutions

Submission deadline: 11 September 2026, 23:59

1. Short answers

[10 marks]

(a)

For $x \geq 0$,

$$p(x) = \exp(ax + b) = e^b e^{ax}, \quad a < 0.$$

Because a probability density must integrate to one,

$$1 = \int_0^\infty e^b e^{ax} dx = e^b \left(-\frac{1}{a} \right),$$

so $e^b = -a$. Letting $\lambda = -a > 0$ gives $b = \log \lambda$ and

$$\boxed{p(x) = \lambda e^{-\lambda x}, \quad x \geq 0.}$$

Thus $p(x)$ is the **exponential distribution** with rate $\lambda = -a$.

(b)

Yes. The categorical latent variable selects one of three exponential components. Marginalizing out z gives

$$\boxed{p(x) = 0.5 \text{Exp}(x \mid 7.8) + 0.4 \text{Exp}(x \mid 0.6) + 0.1 \text{Exp}(x \mid 4.5).}$$

Hence $p(x)$ is a finite mixture model. Equivalently, for $x \geq 0$,

$$p(x) = 0.5(7.8)e^{-7.8x} + 0.4(0.6)e^{-0.6x} + 0.1(4.5)e^{-4.5x}.$$

(c)

Yes. Write

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Then $R^2 = 1 - \text{SSE}/\text{SST}$. Therefore,

$$\boxed{R^2 < 0 \iff \text{SSE} > \text{SST}.}$$

This means that the model predicts the responses worse, in squared-error terms, than the constant baseline $\hat{y}_i = \bar{y}$. It can occur, for example, with a poorly fitted model, a model fitted without an intercept, or evaluation on data different from the fitting data. (Ordinary least squares fitted and evaluated on the same data with an intercept cannot have negative R^2 .)

2. Hidden Markov model

[25 marks]

For a two-step sequence, the HMM factorization is

$$p(z[1], z[2], x[1], x[2]) = \pi(z[1]) p(x[1] | z[1]) T_{z[1], z[2]} p(x[2] | z[2]).$$

(a)(i) Joint probabilities when states are observed

For $x[1] = o_2$ and $x[2] = o_1$, the four possible latent-state paths have the following joint probabilities:

$z[1]$	$z[2]$	$p(z[1], z[2], x[1] = o_2, x[2] = o_1)$
s_1	s_1	$\frac{1}{2}(1-p)ap$
s_1	s_2	$\frac{1}{2}(1-p)(1-a)q$
s_2	s_1	$\frac{1}{2}(1-q)(1-b)p$
s_2	s_2	$\frac{1}{2}(1-q)bq$

(a)(ii) Likelihood when states are hidden

The likelihood is obtained by summing the four joint probabilities over all possible state paths:

$$\begin{aligned} p(o_2, o_1) &= \sum_{z[1], z[2]} p(z[1], z[2], o_2, o_1) \\ &= \frac{1}{2} \left[(1-p)ap + (1-p)(1-a)q \right. \\ &\quad \left. + (1-q)(1-b)p + (1-q)bq \right]. \end{aligned}$$

Thus

$$p(o_2, o_1) = \frac{1}{2} [(1-p)\{ap + (1-a)q\} + (1-q)\{(1-b)p + bq\}].$$

(b)(i) MLE when states are observed

Across the four independent sequences, the transition counts are

$$N_{11} = 1, \quad N_{12} = 1, \quad N_{21} = 0, \quad N_{22} = 2.$$

The factors involving the known emission probabilities p and q , and the fixed initial probabilities, do not affect optimization over a, b . The relevant likelihood is therefore

$$L(a, b) \propto a^{N_{11}} (1-a)^{N_{12}} (1-b)^{N_{21}} b^{N_{22}} = a(1-a)b^2.$$

Equivalently, each transition probability is estimated by its observed relative frequency:

$$\hat{a} = \frac{N_{11}}{N_{11} + N_{12}} = \frac{1}{2}, \quad \hat{b} = \frac{N_{22}}{N_{21} + N_{22}} = 1.$$

Hence

$$\hat{a}_{\text{MLE}} = \frac{1}{2}, \quad \hat{b}_{\text{MLE}} = 1.$$

(b)(ii) MLE when only one emission sequence is observed

Let $\Delta = p - q \neq 0$. From part (a)(ii),

$$\begin{aligned} L(a, b) &= \frac{1}{2} \left[(1-p)\{q + a(p-q)\} + (1-q)\{p + b(q-p)\} \right] \\ &= \frac{1}{2} \left[(1-p)(q + a\Delta) + (1-q)(p - b\Delta) \right]. \end{aligned}$$

This likelihood is affine in both a and b , so its maximum over $0 \leq a, b \leq 1$ occurs at a boundary.

- If $p > q$, then $\Delta > 0$. The coefficient of a is positive and the coefficient of b is negative. Therefore

$$\hat{a}_{\text{MLE}} = 1, \quad \hat{b}_{\text{MLE}} = 0.$$

- If $p < q$, then $\Delta < 0$. The coefficient of a is negative and the coefficient of b is positive. Therefore

$$\hat{a}_{\text{MLE}} = 0, \quad \hat{b}_{\text{MLE}} = 1.$$

These maximizers are unique under the usual nondegenerate assumption $0 < p, q < 1$. At an emission-probability boundary, one parameter may be unidentified: if $p = 1 > q$, the likelihood is independent of a ; if $p < q = 1$, it is independent of b .

(c) Five hidden-state sequences of length 20

Use the **Baum–Welch algorithm**, which is the expectation-maximization (EM) algorithm specialized to HMMs. The latent state paths are unobserved, and enumerating all 2^{20} paths for each sequence is unnecessary.

For each EM iteration:

1. **E-step:** run the forward–backward algorithm separately on each sequence to compute the posterior expected numbers of transitions $s_1 \rightarrow s_1$, $s_1 \rightarrow s_2$, $s_2 \rightarrow s_1$, and $s_2 \rightarrow s_2$. Sum these expected counts over all five sequences.
2. **M-step:** normalize the expected transition counts:

$$a^{\text{new}} = \frac{\mathbb{E}[N_{11}]}{\mathbb{E}[N_{11} + N_{12}]}, \quad b^{\text{new}} = \frac{\mathbb{E}[N_{22}]}{\mathbb{E}[N_{21} + N_{22}]}.$$

Repeat until the data log likelihood converges. Forward–backward evaluates the required posterior quantities in time linear in sequence length (for a fixed two-state model), and each EM iteration does not decrease the observed data likelihood. Because EM can converge to a local optimum, several initial values for (a, b) may be tried and the solution with the largest likelihood retained.

3. Device time to failure**[15 marks]**

This question is mandatory for EE6497 and bonus for IE4497.

Let n_k be the number of observations equal to k . Then

$$\sum_{k=1}^{r+1} n_k = n.$$

(a) Log likelihood

Independence gives

$$\begin{aligned} L(\theta) &= \prod_{k=1}^r \left[\theta^{k-1} (1-\theta) \right]^{n_k} (\theta^r)^{n_{r+1}} \\ &= \theta^{\sum_{k=1}^r (k-1)n_k + r n_{r+1}} (1-\theta)^{\sum_{k=1}^r n_k}. \end{aligned}$$

Therefore the log likelihood is

$$\ell(\theta) = \left[\sum_{k=1}^r (k-1)n_k + r n_{r+1} \right] \log \theta + \left(\sum_{k=1}^r n_k \right) \log(1-\theta).$$

Using $\sum_{k=1}^{r+1} n_k = n$, this can be written more compactly as

$$\ell(\theta) = \left(\sum_{k=1}^{r+1} k n_k - n \right) \log \theta + (n - n_{r+1}) \log(1-\theta).$$

(b) MLE of θ

Define

$$A = \sum_{k=1}^{r+1} k n_k - n, \quad B = n - n_{r+1}.$$

For an interior solution,

$$\frac{d\ell}{d\theta} = \frac{A}{\theta} - \frac{B}{1-\theta} = 0.$$

Thus

$$A(1-\theta) = B\theta \implies \theta = \frac{A}{A+B}.$$

Moreover,

$$\frac{d^2\ell}{d\theta^2} = -\frac{A}{\theta^2} - \frac{B}{(1-\theta)^2} < 0$$

whenever $A, B > 0$, so this stationary point is the unique maximum. Finally,

$$\begin{aligned} A+B &= \left(\sum_{k=1}^{r+1} k n_k - n \right) + (n - n_{r+1}) \\ &= \sum_{k=1}^{r+1} k n_k - n_{r+1}. \end{aligned}$$

Hence

$$\hat{\theta}_{\text{MLE}} = \frac{\sum_{k=1}^{r+1} k n_k - n}{\sum_{k=1}^{r+1} k n_k - n_{r+1}}.$$

Boundary cases. If every observation is $r+1$, the formula gives the boundary MLE $\hat{\theta} = 1$. If every observation is 1, it gives 0; under the stated strict condition $\theta > 0$, this is a supremum as $\theta \downarrow 0$ rather than an attained MLE.